

Akhilesh Vangala

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EDUCATION

New York University

Master of Science, Data Science; GPA: 4.0

Coursework: Machine Learning, Big Data, Advanced Programming for Data Science, Optimization, Computational Linear Algebra

New York, NY

Aug 2025 – May 2027

Gandhi Institute of Technology and Management

Bachelor of Technology, Computer Science with AI/ML; GPA: 3.9

Coursework: Probability, Statistics, Database Management Systems, Algorithms, Big Data Analytics, Machine Learning

India

Aug 2021 – Jul 2025

WORK EXPERIENCE

NYU Rory Meyers College of Nursing

Graduate Research Assistant – Predictive Modeling & Statistical Analysis

- Translated a complex biological problem into a well-scoped technical solution by extending the HiFiMAP genome-wide pipeline, replacing the linear mixed model null with a Cox proportional hazards model and simulating time-to-event data for 5,000 subjects.
- Applied advanced statistical methods including REML-based variance component estimation on a $5,000 \times 5,000$ sparse kinship matrix, validating model outputs against known distributions and confirming residual correlation reached 1.0.
- Built fast variance-component score testing pipelines across 254,145 SNP positions, generating actionable association results and delivering clear reports and findings to research stakeholders.

New York, NY

Apr 2026 – Present

Hochwagen Lab, New York University

Graduate Research Assistant – Data Pipelines, Hypothesis Testing & R&D

- Engineered large-scale data pipelines processing ~1B reads across 54 sequencing samples, applying validation, filtering, and statistical analysis patterns relevant to high-volume behavioral and event data, as part of a global research and development collaboration.
- Identified and resolved a 40x systematic flaw in standard tools through rigorous data validation, designing a strand-aware parser with artifact filtering and true-depth allele frequencies, increasing valid calls by 67x.
- Automated hypothesis testing (Mann-Whitney U), FDR correction, phenotype correlation analysis, and publication-ready visualizations, delivering clear reports and actionable findings to a multidisciplinary team of stakeholders.

New York, NY

Jan 2026 – Present

Dascase Technologies Inc.

Software Engineer Intern – Analytics, Data Models & Platform Services

- Translated business problems into technical solutions by building Python, FastAPI, and SQL-backed services that automated data ingestion, validation, and reporting workflows for internal and client-facing stakeholders.
- Developed data validation checks, anomaly detection scripts, and periodic reports on platform behavior metrics; contributed to existing models and pipelines, enabling cross-functional teams to make data-driven decisions.
- Improved platform data models and ETL pipelines through modular architecture and performance-focused iteration, reducing repeated manual workflows and improving data reliability across analytics and platform services.

Hyderabad, India

May 2024 – Jul 2024

PROJECTS

Customer Behavior & Fraud Intelligence Platform | Python, SQL, Kafka, PyTorch Geometric, SHAP, FastAPI, Redis | Link

- Architected a real-time customer behavior analytics and anomaly detection pipeline using GCN, GraphSAGE, and GAT models to model interaction networks, surface behavioral patterns, and generate risk scores across 500K+ events/day.
- Delivered predictive modeling with explainable inference via SHAP, adaptive thresholding, and strategic insight reports for decision makers – achieving 94% accuracy, 85% false-positive reduction, and sub-50ms latency.

Metrik AI: Large-Scale Behavioral Time-Series Analytics | Python, PySpark, LightGBM, Statistical Modeling, SQL | Link

- Built end-to-end data pipelines over 20.2M time-series behavioral records across 1,449 entities using PySpark, applying feature engineering, data validation, lag features, rolling windows, and hypothesis testing to surface usage patterns.
- Developed predictive models and anomaly rankings for stakeholders – improving RMSE by 23.8% over baselines and producing a 964-entry ranked insight list to drive strategic, evidence-based decisions.

Policy Eviction Analytics Pipeline | Python, SQLite, Statistical Analysis, Automated Reporting | Link

- Built a reproducible analytics pipeline with schema validation, severity-tiered anomaly detection, statistical summaries, and 30 SQL analysis queries across demographic, geographic, and behavioral data to identify actionable patterns.
- Automated stakeholder-facing HTML report generation with A–F data quality grades and derived data models, demonstrating ability to translate analytical findings into clear written communication for non-technical audiences.

SKILLS

- Data Analytics & Statistics:** Customer Behavior Analytics, Statistical Methods, Distributions, Hypothesis Testing, Predictive Modeling, Data Validation, Anomaly Detection, A/B Experiment Analysis, Strategic Insight Reports, Stakeholder Presentations, Business-to-Technical Problem Framing, Research & Development
- Machine Learning & Data Science:** Python, SQL, R, PyTorch, TensorFlow, XGBoost, LightGBM, SHAP, Time-Series Modeling, Graph Neural Networks, Transformers, Machine Learning Applications, Model Evaluation, ML Pipelines
- Data Engineering & Big Data:** PySpark, Kafka, Redis, FastAPI, SQLite, ETL Pipelines, Data Pipelines, Data Products, Data Models, Cloud Computing, Big Data Technologies, Git, Linux, Collaborative Teamwork, Problem-Solving, Technical Writing